CStats - Lab #10: Auto-Encoders

May 12, 2026

Reminder

An auto-encoder uses an encoder and a decoder:

$$x \xrightarrow{\text{encoder } f_\theta} z \xrightarrow{\text{decoder } g_\phi} \hat{x}.$$

The training target is the input itself. A common objective is

$$\min_{\theta,\phi} \frac{1}{n} \sum_{i=1}^{n} \|x_i - g_\phi(f_\theta(x_i))\|_2^2.$$

The latent vector z is also called an embedding, representation, code, or bottleneck.

Exercise 1.

Consider an auto-encoder for input vectors $x \in \mathbb{R}^6$. The encoder maps the input to a 2-dimensional latent code, and the decoder maps the code back to a 6-dimensional reconstruction:

$$z = f_\theta(x) = \sigma(W_e x + b_e), \qquad \hat{x} = g_\phi(z) = \sigma(W_d z + b_d).$$

Here σ is applied element by element.

(a) What are the shapes of W_e, b_e, W_d, and b_d?

(b) Write the reconstruction loss for one data point.

(c) Write the average reconstruction loss for a dataset $x_1, \ldots, x_n$.

(d) Why is this called self-supervised learning?

(e) Why can the latent z be useful for downstream tasks?

(f) Why is the latent z not robust for downstream tasks?

Exercise 2.

This exercise shows a very small bottleneck by hand. Suppose the input is two-dimensional, $x = (x_1, x_2)$, and the auto-encoder uses a one-dimensional code:

$$z = \frac{x_1 + x_2}{2}, \qquad \hat{x} = (z, z).$$

Use the squared reconstruction loss

$$L(x) = \|x - \hat{x}\|_2^2.$$

Consider the three data points

$$x^{(A)} = (3, 3), \qquad x^{(B)} = (4, 0), \qquad x^{(C)} = (0, 4).$$

(a) Compute z and $\hat{x}$ for each data point.

(b) Compute the reconstruction loss for each data point.

(c) Which point is reconstructed perfectly?

(d) What kind of data pattern does this simple auto-encoder preserve?

Exercise 3.
Now consider the smallest possible linear auto-encoder:

$$z = w_e x, \qquad \hat{x} = w_d z, \qquad L = \frac{1}{2}(\hat{x} - x)^2.$$

Here x, z, and $\hat{x}$ are scalars.

(a) Write $\hat{x}$ directly in terms of w_e, w_d, and x.

(b) Derive $\frac{\partial L}{\partial w_d}$.

(c) Derive $\frac{\partial L}{\partial w_e}$.

(d) Use $x = 2$, $w_e = 0.5$, $w_d = 0.4$, and learning rate $\eta = 0.1$. Compute one gradient descent update for w_e and w_d.

Exercise 4.

A de-noising auto-encoder receives a corrupted input but is trained to reconstruct the clean input. Suppose the clean input is

$$x = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

and the corrupted input is

$$\tilde{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

After feeding $\tilde{x}$ through the auto-encoder, the model outputs

$$\hat{x} = \begin{bmatrix} 0.9 \\ 0.7 \\ 0.1 \\ 0.2 \end{bmatrix}.$$

Use mean squared error.

(a) In a de-noising auto-encoder, should the loss compare $\hat{x}$ with x or with $\tilde{x}$?

(b) Compute the de-noising reconstruction loss.

(c) Compute the loss that would be obtained if we incorrectly compared $\hat{x}$ with $\tilde{x}$.

(d) Explain why adding noise during training can produce more robust features.

Exercise 5.
An auto-encoder is trained only on normal data. At test time, the reconstruction error is used as an anomaly score:

$$s(x) = \mathrm{MSE}(x, \hat{x}) = \frac{1}{d}\sum_{j=1}^{d}(x_j - \hat{x}_j)^2.$$

A point is classified as an anomaly if $s(x) > T$, where $T = 0.05$.

Consider the two test points below:

$$x^{(1)} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \hat{x}^{(1)} = \begin{bmatrix} 0.95 \\ 0.90 \\ 0.05 \\ 0.10 \end{bmatrix},$$

$$x^{(2)} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \hat{x}^{(2)} = \begin{bmatrix} 0.40 \\ 0.20 \\ 0.70 \\ 0.30 \end{bmatrix}.$$

(a) Compute the anomaly score for each test point.

(b) Decide whether each test point is normal or anomalous. Briefly explain the reasoning.